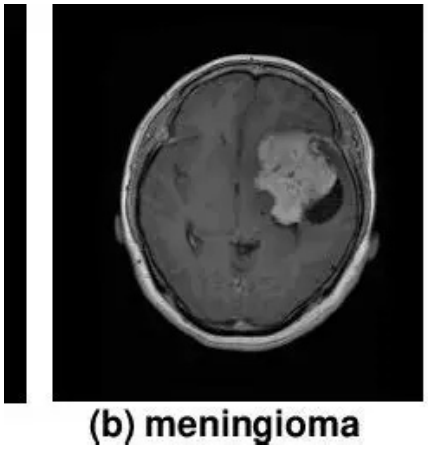
NEURO-AI DIAGNOSTIC CLINICAL REPORT

METAHEURISTIC OPTIMIZED AI
Confidential Medical Document

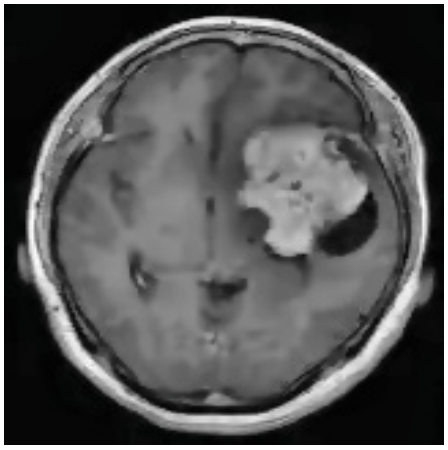
Patient ID: PAT-84920	**Patient Name:** Jane Doe	**Scan Date:** 2026-08-27 08:22:53
Modality: Brain MRI (T1-CE / Axial)	**AI Engine:** Metaheuristic Multi-CNN	**Architecture:** ENSEMBLE

PATHOLOGICAL FINDING: MENINGIOMA DETECTED (Confidence: 27.3%)

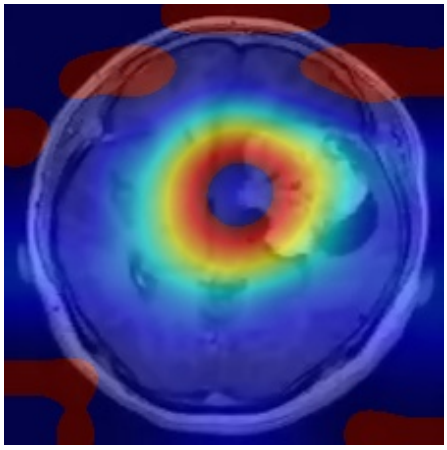
MULTI-STAGE MRI VISUALIZATION & EXPLAINABILITY (XAI)



1. Raw MRI Acquisition
Unprocessed input scan



2. Skull-Stripped & CLAHE
Contour cropped & denoised



3. Grad-CAM++ Saliency
AI Attention Tumor Map

DIAGNOSTIC PROBABILITY DISTRIBUTION & TUMOR LOCALIZATION

Pathology / Class	Probability	Clinical Risk
Glioma	**24.0%**	High
Meningioma	**27.3%**	Moderate
Pituitary Tumor	**25.1%**	Moderate
Healthy / No Tumor	**23.6%**	None

Quantitative Lesion Metric	Observation Value
Estimated Tumor Area Burden	**11.26%** of brain tissue
Anatomical Localization	**Sellar / Central Midline**
Mass Effect Index	**Moderate Mass Effect**
Pathological Description	A slow-growing tumor arising from the arachnoid cells of the meninges. Characterized by well-demarcated margins, dural tail sign, and homogenous contrast enhancement.

Attending Radiologist / Clinician Sign-Off:

Signature & Stamp

AI Assistance Disclaimer:
This report is generated by a deep learning metaheuristic framework as a clinical decision support tool. Final clinical diagnosis must be verified by a board-certified radiologist.